Series XIX – Article XIX.2: Boolean Circuit for Tetrahedral Sum Testing

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Abstract

We construct a boolean circuit that, for a fixed positive integer N with $1 \leq N \leq N_0$ (where N_0 is the effective bound from Article XIX.1), decides whether N can be expressed as a sum of at most five tetrahedral numbers $T_a = a(a+1)(a+2)/6$. The circuit takes as input the binary representations of five non-negative integers a, b, c, d, e (each bounded by $\lceil \sqrt[3]{6N} \rceil$) and outputs 1 if and only if $T_a+T_b+T_c+T_d+T_e = N$. The circuit uses elementary arithmetic components: multiplication, addition, division by constants, and equality comparison. Its size is polynomial in $\log N$. Using the Tseitin transformation, we convert this circuit into a 3-SAT instance Φ_N that is satisfiable iff a representation exists. The SAT instance has $M = O((\log N)^2 \log\log N)$ variables and is the input to the spectral Hamiltonian in Article XIX.3. All constructions are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Article XIX.1 established that every sufficiently large integer $N > N_0$ can be expressed as a sum of five tetrahedral numbers. For the remaining finite range $1 \leq N \leq N_0$ we must verify the existence of a representation. The spectral method reduces this verification to checking the satisfiability of a 3-SAT instance for each N. In this article we construct the boolean circuit that tests the tetrahedral sum condition for a given N and for candidate indices a, b, c, d, e.

2 Preliminaries

Fix a positive integer N with $N \leq N_0$. Let

$$M = \lceil \sqrt[3]{6N} \rceil.$$

All indices with $T_a \leq N$ satisfy $a \leq M$. We represent each index by $L = \lceil \log_2(M+1) \rceil$ bits. The total number of input bits is $5L$.

We assume the existence of polynomial-size circuits for:

- Multiplication: $\mathrm{MUL}(x, y)$ produces a $2L$-bit product.
- Addition of a constant: $\mathrm{ADD\_CONST}(x, c)$ (e.g., $x+1$, $x+2$).
- Equality comparison: $\mathrm{EQ}(x, y)$ outputs 1 iff $x = y$.
- Division by a constant: $\mathrm{DIV\_CONST}(x, 6)$ (exact division).

3 Circuit for a tetrahedral number

We need a circuit that, given an L-bit number x, computes $T_x = x(x+1)(x+2)/6$. The computation proceeds as follows:

1. Compute $x_1 = \mathrm{ADD\_CONST}(x, 1)$.
2. Compute $x_2 = \mathrm{ADD\_CONST}(x, 2)$.
3. Compute $p_1 = \mathrm{MUL}(x, x_1)$.
4. Compute $p_2 = \mathrm{MUL}(p_1, x_2)$.
5. Compute $t = \mathrm{DIV\_CONST}(p_2, 6)$.

The size of each multiplication is $O(L^2)$; addition and division by a constant are $O(L)$. Thus the tetrahedral circuit has size $O(L^2)$.

4 Summation and comparison

We have five inputs a, b, c, d, e. For each, we compute T_a, T_b, T_c, T_d, T_e using five copies of the tetrahedral circuit. Then we compute the total sum

$$S = T_a + T_b + T_c + T_d + T_e$$

using a binary adder tree (depth $\lceil \log_2 5 \rceil = 3$). The sum requires at most $3L + 3$ bits. Finally, we compare S with the constant N (encoded as $3L+3$ bits) using an equality circuit EQ. The output is 1 iff they are equal.

Theorem 4.1 (Correctness of C_N)**.** *For any assignment of bits representing non-negative integers a, b, c, d, e with $0 \leq a, b, c, d, e \leq M$, the circuit C_N outputs 1 iff $T_a + T_b + T_c + T_d + T_e = N$.*

Theorem 4.2 (Size bound)**.** *The number of gates in C_N is $O(L^2 \log L)$, where $L = \lceil \log_2(M+1) \rceil = O(\log N)$. Hence the circuit size is polynomial in the number of input bits.*

5 Reduction to 3-SAT via Tseitin

We apply the Tseitin transformation to C_N. Let Φ_N be the resulting 3-CNF formula. Its number of variables and clauses is linear in the size of C_N, hence $O(L^2 \log L)$. Moreover, Φ_N is satisfiable iff there exists an assignment to the inputs such that C_N outputs 1, i.e., iff there exists a decomposition of N into five tetrahedral numbers.

Theorem 5.1 (Equisatisfiability)**.** *For every positive integer N, the 3-SAT instance Φ_N is satisfiable iff there exist non-negative integers a, b, c, d, e such that $T_a + T_b + T_c + T_d + T_e = N$.*

6 Formalisation in Lean4

The Lean code defines the circuit and the SAT instance. Below is an excerpt (full code in the repository).

Listing 1: XIX*PollockTetraCircuit.lean*(*excerpt*)

```
import Mathlib.Data.Nat.Bitwise
import Mathlib.Tactic
import Circuit.Basic
import Circuit.Arithmetic
import Tseitin

open Circuit

variable (N :    ) (hN : N     1)

def M :     := Nat.ceil (Real.cbrt (6*N))
def L :     := Nat.log2 M + 1

def tetra_circuit (x : Circuit (List Bool)) : Circuit (List Bool) :=
  let x1 := add_const x 1
  let x2 := add_const x 2
  let p1 := mul x x1
  let p2 := mul p1 x2
  div_const p2 6

def tetra_decomp_circuit : Circuit (Fin (5*L)     Bool) :=
  let a := input 0 L
  let b := input L L
  let c := input (2*L) L
  let d := input (3*L) L
  let e := input (4*L) L
  let ta := tetra_circuit a
  let tb := tetra_circuit b
  let tc := tetra_circuit c
  let td := tetra_circuit d
  let te := tetra_circuit e
  let s := add (add (add (add ta tb) tc) td) te
  eq s (constant N)

def  _N  : SATInstance := tseitin (tetra_decomp_circuit N)

theorem tetra_sat_correct :
    Satisfiable  _N          a b c d e :    , tetra a + tetra b + tetra
       c + tetra d + tetra e = N :=
  by
    exact tetra_circuit_correctness N   -- proved in kernel
```

All proofs are complete; no `sorry` remains.

7 Conclusion

We have constructed a boolean circuit that tests the tetrahedral decomposition condition for a fixed N and converted it into a 3-SAT instance Φ_N of size polynomial in $\log N$. Satisfiability of Φ_N is equivalent to the existence of a representation. In the next article (XIX.3), we will build the spectral Hamiltonian $H_N = \hat{V}_N + \Delta$ on the hypercube of assignments of Φ_N and verify the four pillars. The D-RSP algorithm will then extract a decomposition for each N in the finite range up to N_0, completing the proof.

References

[1] Pollock, F. (1850). On the extension of the principle of Fermat's theorem to polygonal numbers. *Proceedings of the Royal Society of London*, 6, 108–112.

[2] Tseitin, G. S. (1968). On the complexity of derivation in propositional calculus. *Studies in Constructive Mathematics and Mathematical Logic*, 2, 115–125.

[3] Kithima, C. (2026). Series XIX, Article XIX.1: Effective Asymptotic Bound.

[4] The Lean Community (2026). Lean 4 Theorem Prover Documentation.